

Question				Marking details		Marks Available					
						AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		Mass of bone	32 [g]	(1)					
				Volume of bone	8 [cm ³]	(1)		2	2		2
		(ii)		$\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{32 \text{ (g)}}{8 \text{ (cm}^3\text{)}} \text{ (ecf) (1)}$ $= 4 \text{ [g/cm}^3\text{]} \text{ (1)}$ Answer = 0.47 [g/cm³] for a volume of 68 cm³ ecf award 2 marks		1	1		2	2	2
	(b)			Particles in bone [more] tightly packed or particles in bone have a greater mass (1) so there is more mass in the same volume (1)		1	1		2		
				Question 1 total		2	4	0	6	2	4